

Technology Stack

Date	15 March 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript	Creates a responsive and user-friendly web interface for collecting HDI indicator values, displaying prediction results, charts, and personalized recommendations.
2	Backend / Server-Side	Python 3.x, Flask	Handles user requests, processes input data, validates data, integrates the Random Forest model, generates predictions, and manages communication between frontend and database.
3	Database / Data Storage	SQLite, CSV Dataset	SQLite stores user accounts and prediction history. The CSV dataset is used for training and testing the Random Forest model.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere / Local Flask Server	Used to deploy the Flask web application so users can access the well-being prediction system through a web browser.
5	Version Control & CI/CD	Git, GitHub	Enables version control, collaboration among team members, code backup, and project management throughout development.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib	Used for data preprocessing, model training (Random Forest), evaluation, visualization of results, and saving/loading the trained machine learning model.